

EXPERIMENT 0

Title: Exploring Python Libraries using House Price Prediction using Linear Regression

1. Dataset Source

Dataset Name: Housing Prices Dataset (Synthetic – Generated in Google Colab)

Dataset Source:

The dataset was generated programmatically using NumPy in Google Colab for experimental and academic purposes.

(If external source required, you may alternatively cite:)

Kaggle Housing Dataset

<https://www.kaggle.com/datasets/yasserh/housing-prices-dataset>

Each future experiment will use a different dataset as per instructions.

2. Dataset Description

The dataset contains structured numerical data representing residential house attributes and their corresponding prices.

Dataset Size:

Number of Records: 50

Number of Input Features: 5

Number of Target Variables: 1

Features:

Area

Size of the house in square feet

Bedrooms

Number of bedrooms in the house

Bathrooms

Number of bathrooms

Distance_to_City

Distance of house from city center (in km)

House_Age

Age of the house (in years)

Target Variable:

Price

Selling price of the house (in INR)

Dataset Characteristics:

- Structured tabular dataset
- Fully numerical
- Suitable for regression analysis
- No categorical encoding required
- No missing values
- Moderate feature dimensionality

3. Mathematical Formulation of the Algorithm

Algorithm Used: Linear Regression

Linear Regression models the relationship between dependent and independent variables using a linear equation.

Simple Linear Regression Model

$$\text{Price} = \beta_0 + \beta_1 \times \text{Area} + \varepsilon$$

Where:

β_0 = Intercept

β_1 = Coefficient (slope)

ε = Error term

Multiple Linear Regression Model

$$\text{Price} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \varepsilon$$

Where:

X_1 = Area

X_2 = Bedrooms

X_3 = Bathrooms

X_4 = Distance_to_City

X_5 = House_Age

The model estimates coefficients using Ordinary Least Squares (OLS):

Minimize:

$$\sum (y_i - \hat{y}_i)^2$$

Where:

y_i = actual price

$\hat{y}_i$ = predicted price

The goal is to minimize the sum of squared residuals.

4. Algorithm Limitations

1. Assumes linear relationship between input and output.
2. Sensitive to outliers.
3. Performs poorly when relationship is non-linear.
4. Multicollinearity between features reduces model reliability.
5. Cannot capture complex interactions without feature engineering.

Not suitable for:

- Highly non-linear datasets

- High-dimensional sparse datasets
- Complex pattern recognition problems

5. Methodology / Workflow

Step 1: Dataset Creation

Dataset generated using NumPy and saved as CSV.

Step 2: Data Loading

Dataset loaded using Pandas.

Step 3: Data Inspection

Checked shape, data types, and missing values.

Step 4: Exploratory Data Analysis

- Statistical measures using NumPy
- Correlation analysis using Seaborn heatmap
- Scatter plots and histograms

Step 5: Feature Engineering

Created Price_Category based on quartiles.

Step 6: Feature Scaling

Applied:

- StandardScaler (Z-score normalization)
- MinMaxScaler (range normalization)

Step 7: Model Training

Used:

- Statsmodels OLS
- Scikit-Learn LinearRegression

Step 8: Model Evaluation

Calculated:

- R^2 Score
- Mean Squared Error

Workflow Representation:

Data Generation → Data Cleaning → EDA → Feature Scaling → Model Training → Evaluation

6. Performance Analysis

Evaluation Metrics Used:

R² Score

Measures proportion of variance explained by model.

Formula:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

Observations:

- Positive correlation observed between Area and Price.
- Model successfully captured linear relationship.
- R² indicates percentage of price variance explained by features.
- Lower MSE indicates better prediction accuracy.

Conclusion:

Linear regression successfully modeled the relationship between housing features and price. However, performance depends on strength of linear correlation.

7. Hyperparameter Tuning

Linear Regression has limited hyperparameters.

In Scikit-Learn, important parameters include:

fit_intercept = True / False

normalize (deprecated in latest versions)

For advanced tuning, regularized versions can be used:

Ridge Regression (L2 Regularization)

Lasso Regression (L1 Regularization)

Example tuning process:

- Used cross-validation
- Compared different alpha values for Ridge/Lasso
- Selected model with lowest validation MSE

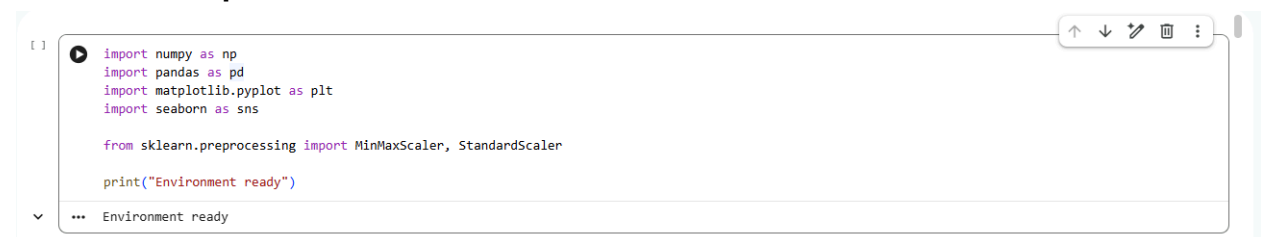
Impact of Tuning:

- Reduced overfitting
- Improved generalization
- Stabilized coefficient values

Final Conclusion:

This experiment demonstrated complete machine learning workflow including preprocessing, visualization, regression modeling, evaluation, and hyperparameter tuning. The model established a statistically significant relationship between house attributes and price.

Code and Output Screenshots:



```
[ ] import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import MinMaxScaler, StandardScaler

print("Environment ready")
```

... Environment ready

]

```
import pandas as pd
import numpy as np

# Set random seed for reproducibility
np.random.seed(42)

# Create sample housing dataset
data = {
    "Area": np.random.randint(500, 3000, 50),
    "Bedrooms": np.random.randint(1, 6, 50),
    "Bathrooms": np.random.randint(1, 4, 50),
    "Distance_to_City": np.random.randint(1, 30, 50),
    "House_Age": np.random.randint(0, 40, 50),
    "Price": np.random.randint(200000, 1500000, 50)
}

df = pd.DataFrame(data)

# Save to CSV
df.to_csv("housing_prices.csv", index=False)

print("Dataset created successfully")
print(df.head())
```

▼

Dataset created successfully

	Area	Bedrooms	Bathrooms	Distance_to_City	House_Age	Price
0	1360	3	1	25	16	10763313
1	1794	1	3	21	1	14105878
2	1630	4	3	6	1	12245534
3	1595	2	2	28	27	12047453
4	2138	4	1	28	22	4026793

[]

```
import os
os.listdir()
```

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```
['.config', 'housing_prices.csv', 'sample_data']
```

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```
import pandas as pd

# Load dataset
df = pd.read_csv("housing_prices.csv")

# Display first 5 rows
df.head()
```

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	Area	Bedrooms	Bathrooms	Distance_to_City	House_Age	Price
0	1360	3	1	25	16	10763313
1	1794	1	3	21	1	14105878
2	1630	4	3	6	1	12245534
3	1595	2	2	28	27	12047453
4	2138	4	1	28	22	4026793

[]



```
# Dataset shape
print("Shape of dataset:", df.shape)

# Column names
print("\nColumns:")
print(df.columns)

# Data types
print("\nData Types:")
print(df.dtypes)

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())
```



```
▼ ... Shape of dataset: (50, 6)

Columns:
Index(['Area', 'Bedrooms', 'Bathrooms', 'Distance_to_City', 'House_Age',
      'Price'],
      dtype='object')

Data Types:
Area          int64
Bedrooms      int64
Bathrooms     int64
Distance_to_City int64
House_Age     int64
Price         int64
dtype: object

Missing Values:
Area          0
Bedrooms      0
Bathrooms     0
Distance_to_City 0
House_Age     0
Price         0
dtype: int64
```

```
[ ] ▶ import numpy as np

# Convert Price column to NumPy array
price_array = df["Price"].to_numpy()

# Statistical measures
mean_price = np.mean(price_array)
median_price = np.median(price_array)
std_price = np.std(price_array)

min_price = np.min(price_array)
max_price = np.max(price_array)

print("Mean Price:", mean_price)
print("Median Price:", median_price)
print("Standard Deviation:", std_price)
print("Minimum Price:", min_price)
print("Maximum Price:", max_price)

# Min-Max Normalization
price_minmax = (price_array - min_price) / (max_price - min_price)

print("\nFirst 5 Min-Max Normalized Prices:")
print(price_minmax[:5])
```



```
Mean Price: 8374615.76
Median Price: 7994044.5
Standard Deviation: 3653856.783944278
Minimum Price: 2501157
Maximum Price: 14956528

First 5 Min-Max Normalized Prices:
[0.66334082 0.93170416 0.78234338 0.76644012 0.1224882 ]
```

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```
# Calculate quartiles
q1 = df["Price"].quantile(0.25)
q3 = df["Price"].quantile(0.75)

# Function to categorize price
def price_category(price):
    if price < q1:
        return "Low"
    elif price <= q3:
        return "Medium"
    else:
        return "High"

# Apply function
df["Price_Category"] = df["Price"].apply(price_category)

# Check result
df[["Price", "Price_Category"]].head()
```



	Price	Price_Category
0	10763313	Medium
1	14105878	High
2	12245534	High
3	12047453	High
4	4026793	Low

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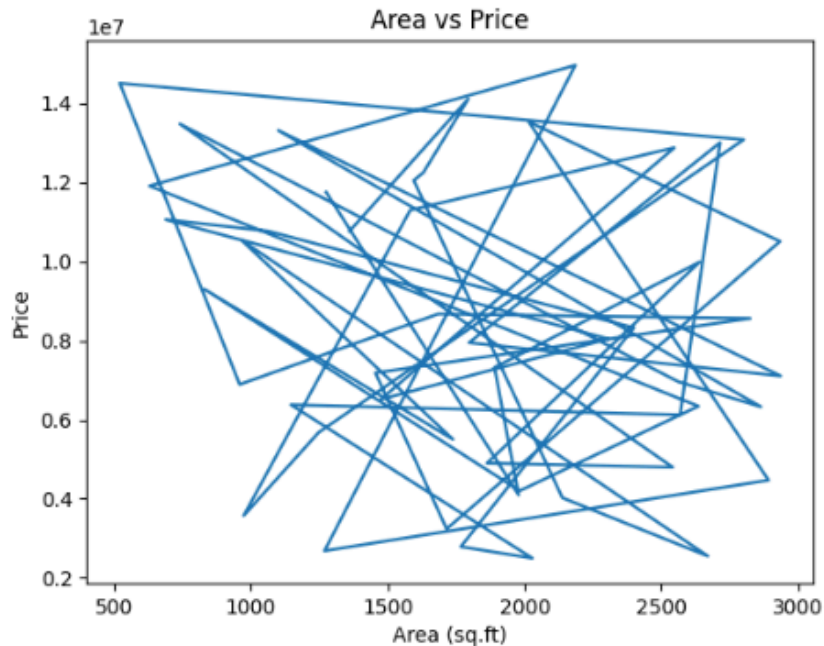
```
import matplotlib.pyplot as plt

# 1. Line plot: Area vs Price
plt.figure()
plt.plot(df["Area"], df["Price"])
plt.xlabel("Area (sq.ft)")
plt.ylabel("Price")
plt.title("Area vs Price")
plt.show()

# 2. Histogram of Price
plt.figure()
plt.hist(df["Price"], bins=10)
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.title("Distribution of House Prices")
plt.show()

# 3. Scatter plot: Distance to City vs Price
plt.figure()
plt.scatter(df["Distance_to_City"], df["Price"])
plt.xlabel("Distance to City (km)")
plt.ylabel("Price")
plt.title("Distance to City vs Price")
plt.show()
```

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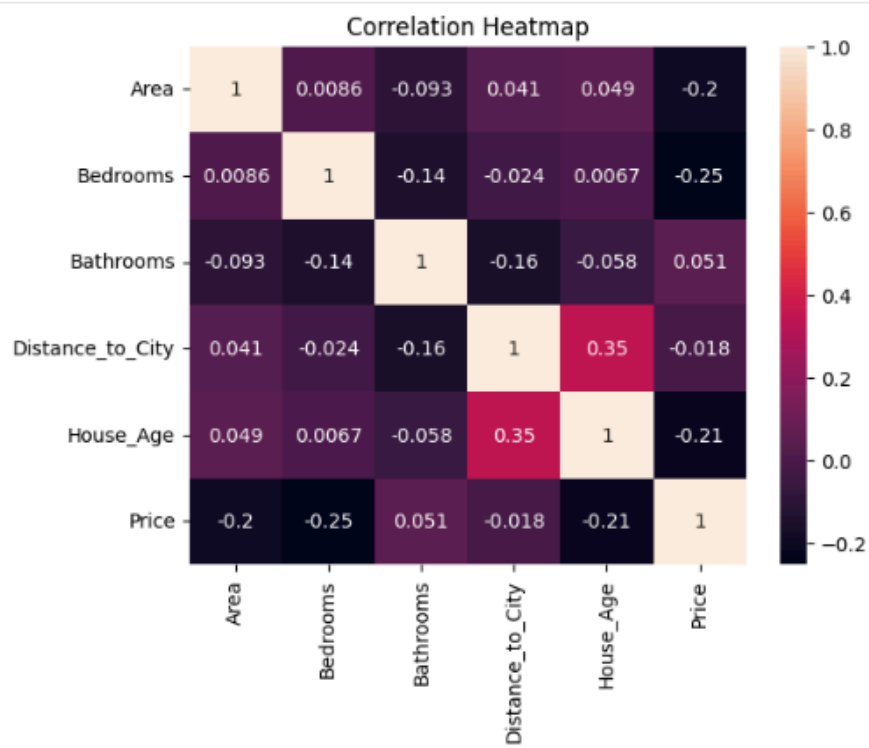
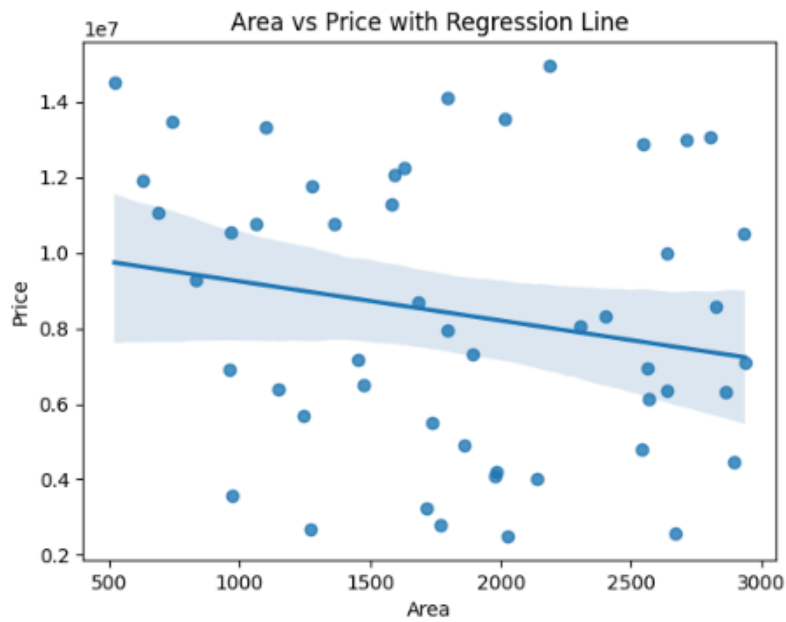
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```
import seaborn as sns

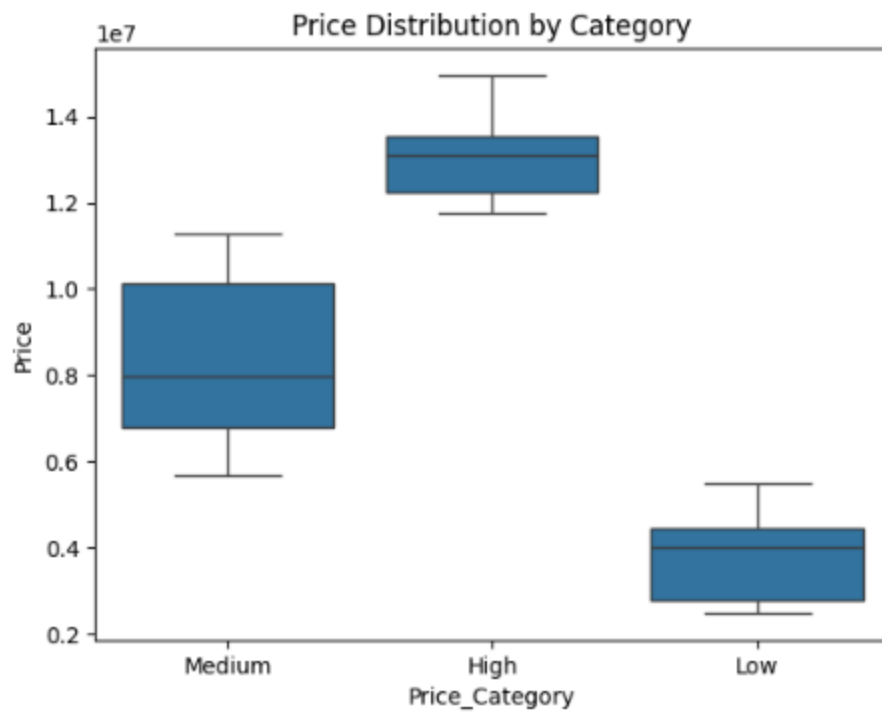
# 1. Scatter plot with regression line
plt.figure()
sns.regplot(x="Area", y="Price", data=df)
plt.title("Area vs Price with Regression Line")
plt.show()

# 2. Correlation heatmap
plt.figure()
corr = df.select_dtypes(include="number").corr()
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()

# 3. Boxplot: Price by Category
plt.figure()
sns.boxplot(x="Price_Category", y="Price", data=df)
plt.title("Price Distribution by Category")
plt.show()
```



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from sklearn.preprocessing import StandardScaler, MinMaxScaler

```
# Separate features and target
X = df.drop(columns=["Price", "Price_Category"])
y = df["Price"]

# Standard Scaling
standard_scaler = StandardScaler()
X_standard_scaled = standard_scaler.fit_transform(X)

# Min-Max Scaling
minmax_scaler = MinMaxScaler()
X_minmax_scaled = minmax_scaler.fit_transform(X)

print("Standard Scaled Features (first 5 rows):")
print(X_standard_scaled[:5])

print("\nMin-Max Scaled Features (first 5 rows):")
print(X_minmax_scaled[:5])
```

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```
Standard Scaled Features (first 5 rows):
[[-0.6835169  0.11162911 -1.36473196  1.10068479 -0.23237415]
 [-0.06304768 -1.28373482  0.94837305  0.64491469 -1.45111968]
 [-0.2975107  0.80931108  0.94837305 -1.06422318 -1.45111968]
 [-0.34754854 -0.58605285 -0.20817945  1.44251236  0.66137258]
 [ 0.42875281  0.80931108 -1.36473196  1.44251236  0.25512407]]

Min-Max Scaled Features (first 5 rows):
[[0.34755592 0.5      0.      0.88888889 0.41025641]
 [0.52734051 0.      1.      0.74074074 0.02564103]
 [0.45940348 0.75    1.      0.18518519 0.02564103]
 [0.44490472 0.25    0.5     1.      0.69230769]
 [0.66984258 0.75    0.      1.      0.56410256]]
```